

Sentinel — Additions for V4

This document specifies the new architectural content to add to the paper, organized by the section each item belongs in. It captures every design decision reached through discussion since V3, with the reasoning each one rests on, so it can be drafted into prose directly. A short list of smaller items still outstanding from the V3 review is included at the end.

1. Tier 3 Message-Delivery Mechanism (Section 3.6, expanding the existing Table 1 row)

This is not a new escalation tier. Table 1 already names the mechanism — *"optionally, with prior consent, an encrypted context message to a pre-designated contact"* — this section specifies how that mechanism actually works.

Generation: pre-authored, not live. No message is composed by Gemma at trigger time. A small, fixed library of templates (2–3 per trigger category) is authored offline and reviewed with the same clinician-input process already applied to the PHQ-9/GAD-7 lexicon in Section 3.3. At trigger time, the system performs two operations only: (1) selects the template matching the signal that fired, and (2) fills a small set of variables — contact's saved name for the user, a coarse tier/category label, the Tele MANAS number. No free text is generated in the moment.

Rationale: extends a principle the paper already applies everywhere else — constrain generation to pre-vetted content wherever the cost of an error is high — to the one channel that didn't yet have it. Also means this channel needs no new live safety layer to validate; the safety work happens once, at authoring time, the same way it does for the CBT-nudge lexicon.

Two template sets, not one. Self-harm and harm-to-others triggers need separate templates — tone and content appropriate for a friend supporting someone in crisis are not appropriate, and may be actively wrong, for a credible-threat-to-a-third-party disclosure.

Content structure. Each template contains: (a) the tier/category, not a verbatim transcript or a generated narrative of "the situation"; (b) a direct instruction for the contact to act; (c) a resource for the *contact*, not only the user — Tele MANAS and comparable lines take calls from a worried third party, not only from the person at risk. Content should be grounded in the QPR (Question–Persuade–Refer) gatekeeper-training model — the standard evidence-based framework for what an untrained layperson should do, which itself terminates at *refer* rather than asking the bystander to improvise an intervention. **Citation needed:** source a primary/peer-reviewed reference for QPR's evidence base before submission, not a training-provider page.

Delivery channel. Text by default. If a call path is included, it is a one-tap call action inside the text — the same pattern Table 1 already uses for Tele MANAS at Tier 2 ("one-tap connection") — so a human places the call, not Sentinel. An autonomous outbound call is not recommended: reliability is uncertain, and a synthesized voice describing a mental health crisis is difficult to distinguish from unwanted robocall traffic.

No retry/escalation-on-no-response logic. Tele MANAS surfacing to the user already fires immediately and unconditionally at Tier 3, independent of the friend channel's outcome — the friend contact is additive, not a single point of failure. Automatic escalation on no response would reintroduce the automated-dispatch pattern Section 3.5 already excludes on principle.

Setup-time preview requirement. Before a user confirms a Tier 3 contact, they see the literal message text that would be sent. This operationalizes the consent-specificity commitment already made in Section 5.5.

2. Friend-Contact Positioning (Section 2.1 / 3.6 framing)

State explicitly that the friend-contact channel operates **alongside** Tele MANAS, not as a substitute or superior alternative — as a stated design principle, not left implicit.

Rationale: QPR's own materials describe trained-layperson gatekeeper response as "a citizen emergency response," explicitly distinguished from treatment. Relational closeness provides immediacy and warmth a national helpline cannot; it does not substitute for crisis-specific training. Avoid any framing suggesting the friend channel outperforms or replaces Tele MANAS.

3. Harm-to-Others: Sharpening the Already-Flagged Open Question (Sections 3.6 and 5.2)

Two additions to the existing open-question treatment, which currently asks only "is Tele MANAS the right channel":

Name the deeper question directly: even granting Tele MANAS is the correct channel, what obligation or action does a duty-to-protect disclosure create for a counsellor who has not spoken with the user, regarding a threatened third party they cannot identify or contact? Routing resolves which phone number receives the alert; it does not resolve what that recipient is positioned to do with it.

Name a new privacy gap: a harm-to-others classification necessarily involves information about a person who never enrolled in Sentinel. Section 5.1's zero-knowledge framing is currently scoped entirely to the user's own data. Name this as open, consistent with how the routing question is already handled.

Note on the Tele MANAS "stepped confidentiality" precedent (Section 6, below): it is a strong anchor for the *self-harm* side of this question — minimum-necessary disclosure only at a defined severity threshold is real, demonstrated Tele MANAS practice. It does not, by itself, resolve the harm-to-others sub-question, since that protocol is built around protecting the caller, not an unidentified third party. Keep both halves of 5.2 explicitly open rather than treating the precedent as a full resolution.

4. Financial and Peer-Pressure Signal (Sections 3.2 / 3.6)

Captured only as explicit, user-typed statements within Sentinel's own first-party text surfaces — the same surfaces Section 3.2 already scopes keystroke-dynamics capture to. State this as explicitly as the IME/AccessibilityService exclusion already is in 3.2.

Surfaces to the Tier 2/3 human reviewer as supplementary context only. Does not enter the Phase 2 automated ensemble or the RAG-grounded generation pipeline.

Rationale: no validated clinical instrument exists for these constructs (unlike PHQ-9/GAD-7), so keeping them out of automated pathways preserves the RAG-groundedness discipline the generation-safety argument in Section 2.4/3.3 depends on.

5. Academic Records (Section 3.2)

Captured only via voluntary, user-initiated upload (grades, marks, achievements) — not institutional-system integration. This is what keeps the feature consistent with the "user is the sole party with access

to their own data" claim in Section 2.1.

Treated the same as the financial/peer-pressure signal: human-facing context at Tier 2/3 only, not an automated input at this stage.

Academic performance has more established published correlation with mental health than peer or financial pressure does, but still falls short of PHQ-9/GAD-7's clinical specificity. Name it as a Stage 0 validation target for future work.

6. Disclosure Avoidance and the Stigma Barrier (new — Introduction, Section 5.2, Section 5.3 or a new 5.x)

The problem to state explicitly: systems that rely on informing a personal contact are structurally ineffective for a specific, well-documented population — people whose primary barrier to help-seeking is fear of being found out, not lack of access. Cite **Clement et al. (2015, *Psychological Medicine*)**: stigma is the fourth-highest-ranked barrier to help-seeking overall in their synthesis, and disclosure concern specifically is the single most frequently reported stigma-related reason for avoidance.

The structural answer already in your architecture, now worth stating as a defended design claim rather than an assumption: a friend/family contact and a Tele MANAS contact are not the same disclosure event. Tele MANAS's anonymity-by-default design — no name, age, gender, or location required; masked caller ID; no call recording — is structurally exempt from the disclosure-cost calculus that makes informal contacts unusable for this population. **Cite the British Journal of Psychiatry (2026), "India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health"** for the anonymity and data-minimization claims specifically — get full bibliographic details before adding to the reference list.

Design additions this motivates:

Reframe the on-device Tier 1 layer (Gemma-based EMA, PHQ-9/GAD-7-grounded) as a *complete, self-sufficient option*, not merely a funnel toward escalation. For the most disclosure-averse users, this is the only tier they will actually engage with, and it should carry zero implication that it needs to lead anywhere. State explicitly, at the point Tele MANAS is surfaced, that the suggestion itself was generated entirely on-device and that calling is the user's own choice — an accurate claim your architecture already earns, worth saying out loud at the moment of highest resistance rather than leaving implicit.

Add an explicit boundary-condition statement to Section 5.7's limitations: no design can compel disclosure from a user who has declined it. Sentinel's contribution is lowering the activation energy of self-initiated help-seeking, not guaranteeing an outcome.

Resolution-statement language (corrected for consistency with the actual Section 3.6 design — do not use an earlier draft that states all escalation routes through Tele MANAS, which would contradict the pre-designated-contact mechanism already specified):

"Sentinel's escalation model does not require disclosure to personal contacts to remain effective. Tele MANAS — reachable at Tier 2 and Tier 3 alike — is structurally exempt from the disclosure-cost calculus that makes a friend or family contact unusable for some users: calls require no name, age, gender, or location, phone numbers are masked, and nothing is recorded. It remains a viable channel for users who decline the optional pre-designated-contact mechanism in Section 3.6. Escalation of either kind is always presented as an offer, never executed without the user's own action. We acknowledge as an explicit design boundary that no passive-sensing system can compel disclosure from a user who has declined it."

7. Target Population — Scoping Recommendation (Introduction / Abstract)

State the realistic target population explicitly rather than relying on the abstract's "over one billion people" framing alone. Tele MANAS's own usage data shows roughly 70% of callers fall in the 18–45 age range, concentrated in working adults and students — matching the population your own recent additions (institutional counselling, academic records, career pressure) are already shaped around. Recommended framing: **Android-owning, Tele MANAS-eligible individuals in India — disproportionately students and early-career adults — experiencing distress not currently being disclosed through any existing channel.** This is a narrower claim than the abstract currently implies, and a more defensible one.

Also still outstanding (from the V3 review, not addressed since)

Item	Where
ProMind-LLM not yet cited — MindGuard's own direct successor, same lab; conspicuous given how thoroughly MindGuard is treated	Section 2.1
"Five-stage pipeline" prose numbering bundles three distinct processes into one stage, inconsistent with Figure 1's separate boxes	Section 3.1
ProKnow's "96% reduction" glossed as "knowledge-capture-violation metric" rather than the source's actual term (averaged squared rank error)	Section 2.4
Confirm Table 1 and the Stage table render as actual tables in the final PDF, not run-together prose	Section 3.5 / 4.3